

Adding Integers: Numberline

Adding Integers on a Number Line

Another way to visualize addition is to add on a number line.

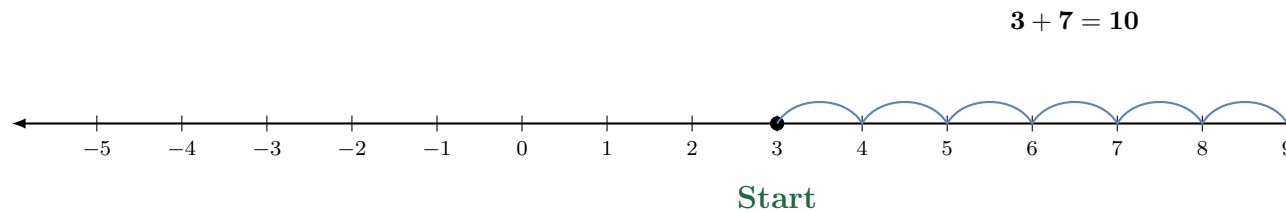
Procedure 1 (Steps for Adding Integers on a Numberline). (a) Locate the first number on the number line.

(b) If the second number is **positive**, move to the **right** that many units.

(c) If the second number is **negative**, take the absolute value of the number and move to the **left** that many units.

Example 1. Use a numberline to find the sum $3 + 7$.

Explanation. Start at 3, then move 7 units to the **right**. Movement is to the right, since a **positive** number is added to 3.



Alt text: Horizontal number line from negative 5 to 10, with tick marks and labels for each integer. There is a solid black dot at 3. The word “Start” is written below the number 3 in green. A series of connected, blue arches move rightward from 3, landing on each integer until stopping with a right-pointing arrow at 10. The word “Stop” is written below the number 10 in orange. The equation “3 plus 7 equals 10” is displayed centered above the arches.

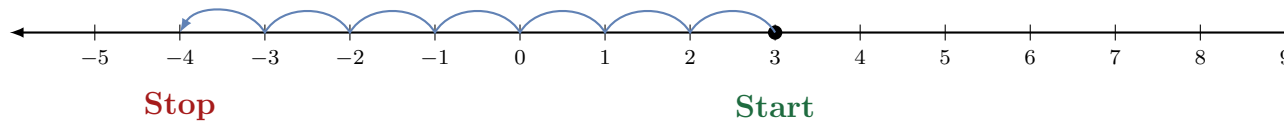
Example 2. Use a numberline to find the sum $3 + (-7)$.

Explanation. Start at 3 then move $|-7| = 7$ units to the **left**. Movement is to the left, since a **negative** number is added to 3.

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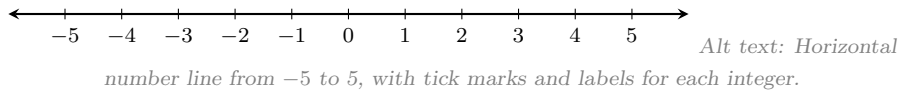
$$3 + (-7) = -4$$



Alt text: Horizontal number line from negative 5 to 10, with tick marks and labels for each integer. There is a solid black dot at 3. The word “Start” is written below the number 3 in green. A series of connected, blue arches move leftward from 3, landing on each integer until stopping with a left-pointing arrow at negative 4. The word “Stop” is written below the number negative 4 in orange. The equation “3 plus negative 7 equals negative 4” is displayed centered above the arches.

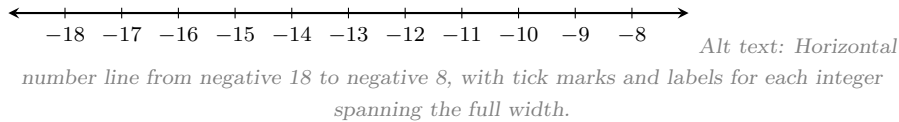
Problem 1 Add using the number line shown below.

$$-2 + 3 = 1$$



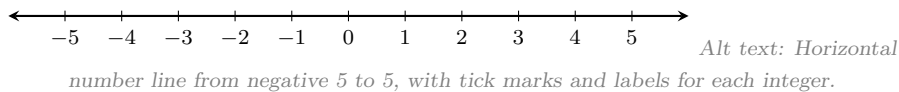
Problem 2 Add using the number line shown below.

$$-12 + (-3) = -15$$



Problem 3 Add using the number line shown below.

$$-1 + 4 = 3$$



Problem 4 Does the answer to the previous problem change if you switch the order of the numbers and find the value of $4 + (-1)$?

Multiple Choice:

- (a) No ✓
- (b) Yes

Feedback(correct): That's right, the answer does NOT change! This is due to a property of addition called the Commutative Property. The **Commutative Property of Addition** states that numbers can be added in any order and the result will be the same. Therefore, $-1 + 4 = 4 + (-1)$.